ABHINANDAN ANIL GAVALI

Bhopal, MP, India | abhinandangavali310@gmail.com | GitHub: abhinandangavali310-byte

EDUCATION

VIT Bhopal University
Integrated M.Tech in Computer Science Engineering (Specialization in Cyber Security) ***CGPA: 8.0 / 10.0***
Expected Graduation: May 2027

Academic Cohort Program Representative, CSE Cyber Security (2022–Present)

RESEARCH PUBLICATIONS

- "Naive Evaluation Overstates Re-Identification Risk in Shared Honeypot Threat Intelligence" accepted at IEEE International Conference on Computer & Applications (IEEE-ICCA). (Evaluated empirical privacy leakage risks in multi-organization threat data feeds using the MURHCAD dataset).
- "AI-Based Deceptive Intelligence System for Proactive Cybersecurity" published at Mantech Publications. (Designed generative adversarial network [GAN] and deep reinforcement learning frameworks to automate cyber deception and counter adversarial reconnaissance).
- “Cybersecurity Essentials” Published technical book focusing on the operational integration of AI tools in cyber security

RESEARCH PROJECTS

1. Adversarial Machine Learning for Intrusion Detection

- Investigated evasion attacks against machine learning-based intrusion detection systems under adversarial conditions.
- Implemented adversarial testing experiments using Python, TensorFlow, and Scikit-learn to stress-test classifier boundaries.
- Evaluated model robustness across diverse perturbation attack scenarios and quantified degradation metrics.

2. AI-Powered Malware Classification Framework

- Developed an automated malware classification prototype extracting comprehensive static and dynamic execution features.
- Applied ensemble classifiers including Random Forest and XGBoost to categorize complex binary payloads with high accuracy.
- Evaluated model performance and generalization across diverse malware families and obfuscation techniques.

3. Privacy-Preserving Threat Intelligence Sharing

- Explored federated learning architectures for collaborative, decentralized threat intelligence exchange between organizations.
- Investigated privacy guarantees, gradient leakage vulnerabilities, and communication overhead in distributed nodes.
- Evaluated the practical feasibility and convergence metrics of distributed cybersecurity analytics models.

TECHNICAL EXPERIENCE & INTERNSHIPS

Cyber Security Intern — Bhopal Police Commissionerate (Madhya Pradesh Police Cyber Cell) Oct 2024 – Mar 2025
Bhopal, India

- Executed digital forensic acquisitions and binary/malware artifact analysis utilizing Autopsy, Volatility, and sandboxing tools on live case data.

- Analyzed regional cybercrime vectors, documented victim behavioral patterns, and synthesized intelligence briefs to support law enforcement investigations.

Defence and Space Tech Intern — Bharat Space Education Research Centre Dec 2025 – Jan 2026
Remote / India

- Investigated vulnerability surfaces in mission-critical aerospace communication pipelines and telemetry protocols.
- Analyzed resilience metrics and cryptographic key exchange mechanisms against strict secure network baselines.

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Core Member, Public Relations (PR) Team — VIT Model United Nations (MUN)

- Managed external institutional communications, delegation outreach, and media relations for major annual Model United Nations conferences.
- Coordinated strategic promotional campaigns and stakeholder engagement across multiple university chapters.

TECHNICAL SKILLS & RESEARCH METHODOLOGY

Programming:	Python, C++, Java, SQL, Bash
Cybersecurity:	Network Security, Vulnerability Assessment, Malware Analysis, Digital Forensics, Incident Response, Adversarial ML
AI/ML Frameworks:	TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy
Security Tools:	Wireshark, Nmap, Burp Suite, Metasploit, Ghidra, Volatility, Suricata, Autopsy
Research & Infra:	Jupyter Notebook, MATLAB, Git, Linux, LaTeX, AWS, Docker, Kubernetes
Methodology:	Experimental research design, Statistical hypothesis testing, Feature engineering, Model evaluation, Reproducible documentation

CERTIFICATIONS & STANDARDIZED TESTS

- Certified Ethical Hacker (CEH v12) (Packt / Coursera) — January 2025
- Junior Cybersecurity Analyst Career Path (Cisco Networking Academy) — Sep 2024 – Jan 2025
- TOEFL Exam Score: 92 / 120